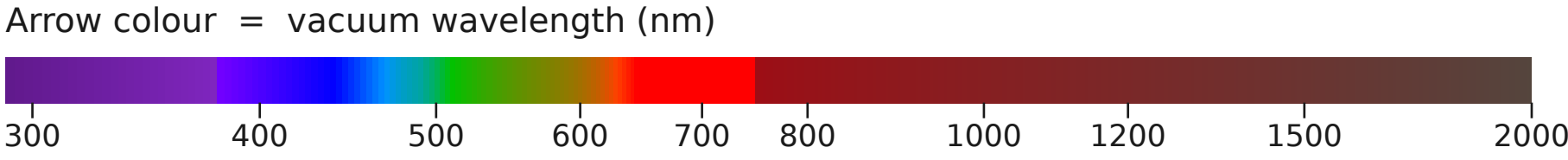


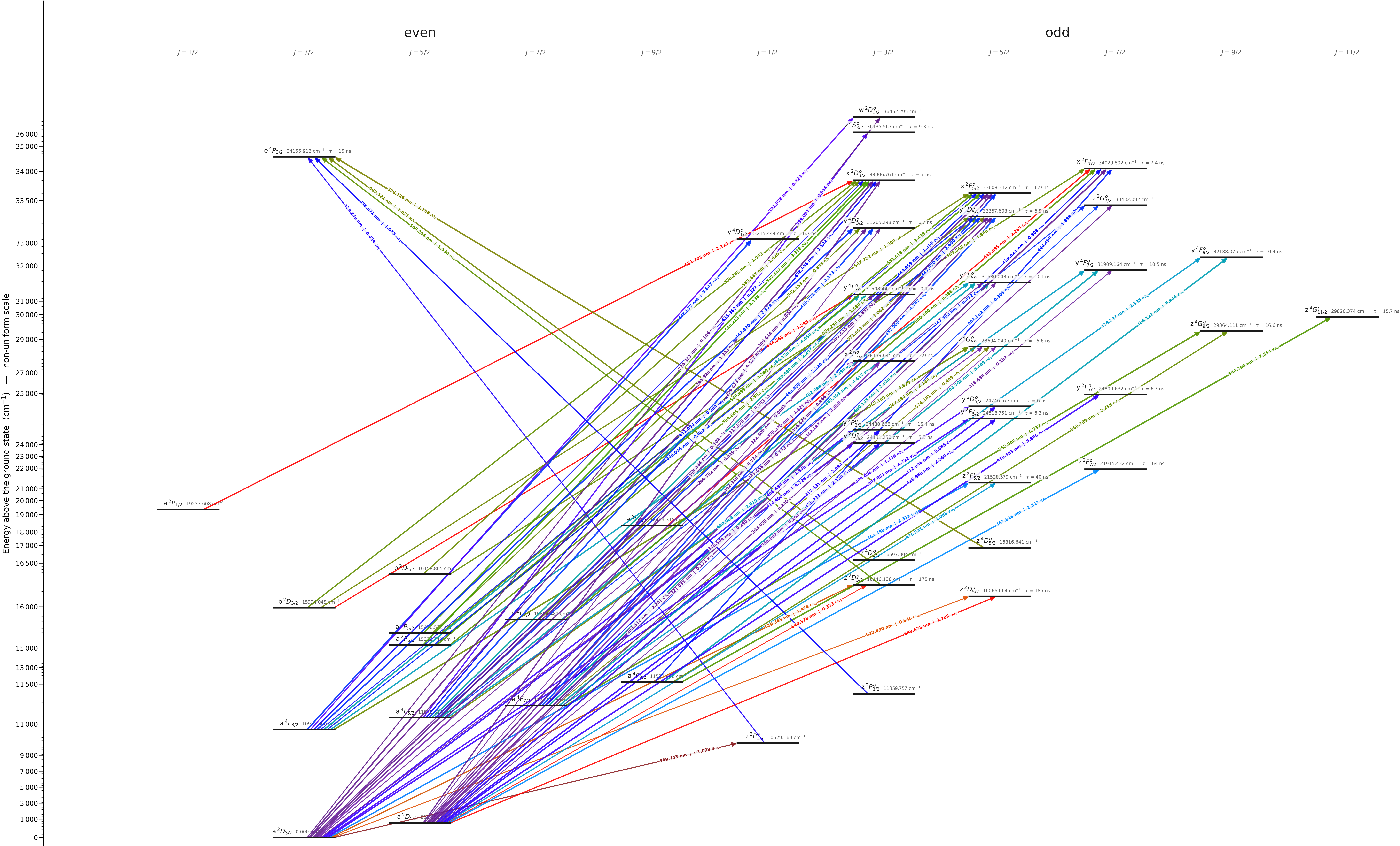
Yttrium

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||e\mathbf{r}||J'\rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Y (Y I). Strongest classified transitions below 2 μm (NIST): 44 levels, 90 lines.

Y⁺ ionisation limit 50145.60 cm^{-1} = 6.21726 eV (199.419 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e\mathbf{r} J'\rangle $ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^2D_{3/2} - y^2F_{5/2}^o$	407.8511	407.7360	735.0537	4.722	1.110e+08	25.26	70	strong 2D-2F resonance line
$a^2D_{5/2} - y^2F_{7/2}^o$	410.3527	410.2369	730.5727	5.886	1.270e+08	23.75	85	strong 2D-2F resonance line
$a^2D_{5/2} - y^2D_{5/2}^o$	412.9463	412.8299	725.9841	5.685	1.550e+08	26.53	93.3	strong 2D-2D resonance line of the ground term
$a^2D_{3/2} - y^2D_{3/2}^o$	414.4004	414.2836	723.4367	4.726	1.590e+08	30.03	84.1	strong 2D-2D resonance line of the ground term
$a^4F_{9/2} - z^4G_{11/2}^o$	546.7983	546.6464	548.2688	7.854	6.370e+07	10.14	100	proposed laser-cooling
$a^2D_{3/2} - z^2P_{1/2}^o$	949.7426	949.4821	315.6565	≈ 1.099	1.429e+06	-	-	proposed type-II MOT / narrow cooling line from the true...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁸⁹Y hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$a^2D_{3/2}$	-57.220	-	F=1: +71.53 F=2: -42.91
$a^2D_{5/2}$	-28.850	-	F=2: +50.49 F=3: -36.06
$z^2D_{5/2}^o$	-216.100	-	F=2: +378.18 F=3: -270.12
$z^2D_{3/2}^o$	88.300	-	F=1: -110.38 F=2: +66.22
$z^4D_{3/2}^o$	-38.300	-	F=1: +47.88 F=2: -28.73
$z^4D_{5/2}^o$	-132.600	-	F=2: +232.05 F=3: -165.75
$e^4P_{3/2}$	-124.800	-	F=1: +156.00 F=2: -93.60

Measured A, B constants; sources in the data file.